

Shram Kadia

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San Jose, CA

SUMMARY

Backend engineer specializing in LLM systems, multi-model orchestration, and cost-optimized AI infrastructure. Experienced building production-grade agentic systems, implementing real-time data integration, and designing cost-aware inference pipelines.

WORK EXPERIENCE

OpenPRA Org

Aug 2025 - Present

Software Engineer

- Owned **end-to-end delivery** of a probabilistic fault-tree generator for nuclear safety analysis, shipped from concept to production in 8 weeks using **Python**, **TypeScript**, **Node.js**, and Docker with full CI/CD automation.
- Authored 25+ Jest unit tests** and CI checks guaranteeing numerical stability and reproducible outputs, maintaining **full coverage** on safety-critical probability calculations.
- Architected **type-safe probability classes** across packages, enabling rapid testing and iteration loops.
- Redesigned XML processing pipeline with lazy-parse techniques, cutting debugging cycles by **40%** and accelerating risk-assessment turnaround.

NC State University

Jul 2024 - Mar 2025

Backend Engineer - Research Assistant

- Designed and deployed a **production RAG stack** using **LangChain**, **FastAPI**, **FAISS**, **HuggingFace** transformers, and Redis, achieving **85% extraction accuracy** and sub-200ms latency under concurrent load.
- Engineered retrieval pipeline with **semantic chunking**, **confidence thresholds**, and exponential-backoff retry logic, eliminating **60% of manual work** and saving the department **30 hours/week**.
- Integrated distributed tracing and structured logging** into the FastAPI service; used telemetry to identify bottlenecks and drive a **30% throughput improvement**.

Resilient Tech

Jan 2023 - May 2023

Software Engineer

- Migrated critical ERP systems** to RESTful **Python APIs** across 5 enterprise rollouts, cutting production downtime by **30%** and sustaining **99% uptime**.
- Resolved **100+ production support tickets** by tracing root causes through front-end and back-end workflows, lifting CSAT by **12%**.
- Built and maintained **client-facing React interfaces** alongside Python backend APIs, delivering full-stack features using **Git-based version control** and Agile workflows.

PROJECTS

Travel Agent ATLAS

Python · FastAPI · LangChain · OpenAI-Compatible Proxy

- Built runtime LLM switching across 4 models with unified OpenAI-compatible proxy.
- Tracked per-request token usage and cost, enabling live cost/performance tradeoffs.
- Fired parallel Exa neural search queries, eliminating retrieval latency from critical path.

Restaurant Voice Hub

Python · FastAPI · PostgreSQL · ElevenLabs

- Architected agentic orchestration with 5 callable tools for autonomous voice ordering.
- Engineered real-time PostgreSQL sync via CSV pipeline, cutting menu update to seconds.
- Deployed multi-tenant architecture across Vercel/Render, ready for scaling.

EDUCATION

M.S. Computer Science, North Carolina State University

May 2025

GPA: 3.63/4

B.Tech Computer Engineering, Pandit Deendayal Energy University

May 2023

GPA: 9.62/10

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI & LLMs: LangChain, FastAPI, Gemini, OpenAI, Qwen

Data & Cloud: PostgreSQL, Redis, FAISS, AWS, GCP, Docker

Tools: Exa Search, Observability, Git, CI/CD